

Technical Design Report: Multi-Sensor Structural Vibration Analyzer

1. Executive Summary

In the domain of structural health monitoring (SHM), high-precision vibration analysis is the cornerstone of life-cycle assessment for critical infrastructure. Modern engineering demands high-fidelity data to detect infinitesimal modal shifts in bridges and industrial assets before they progress to catastrophic failure. This report outlines a hardware architecture that achieves professional-grade results—comparable to costly, proprietary DAQ systems—by utilizing a highly integrated, cost-effective embedded ecosystem. By shifting the paradigm from expensive centralized hardware to distributed, scalable sensing nodes, this architecture democratizes real-time frequency-domain analysis. The system synthesizes the computational power of the ESP32 System-on-Chip (SoC) with an array of MPU-6050 Micro-Electro-Mechanical Systems (MEMS) sensors. To overcome the physical limitations of I2C address bus contention, the design incorporates a TCA9548A/PCA9548A multiplexer, allowing the controller to interface with up to eight independent sensor nodes. The core operational objective is the maintenance of a deterministic, high-speed raw data stream to a host computer. By offloading the computationally intensive Fast Fourier Transform (FFT) to a host environment, the embedded system remains focused on precise temporal alignment and signal integrity, providing a blueprint for modern structural diagnostics.

2. Hardware Architecture & System Integration

The efficacy of a structural analyzer is dictated by its ability to resolve timing at the microsecond level while managing substantial data throughput. Selecting components that balance high-speed internal buffering with robust communication protocols is essential to meet the rigorous sampling requirements of civil engineering applications.

The ESP32 Processing Core

The heart of the system is the ESP32 Series SoC, featuring the Xtensa® 32-bit dual-core LX6 microprocessor. Running at a clock frequency of 240 MHz, this dual-core architecture is strategically utilized through "Core Pinning." In this configuration, the sampling Interrupt Service Routine (ISR) is pinned to Core 1 to ensure deterministic I2C polling, while Core 0 handles the non-deterministic overhead of the Wi-Fi or Bluetooth v4.2 (including BR/EDR and Bluetooth LE) stacks.

- **Memory and Throughput:** The 520 KB on-chip SRAM facilitates large sample buffers, which are essential for preventing data loss during burst transmissions. The system utilizes dedicated memory access pathways and the ESP32's internal bus architecture to move data from the I2C peripheral to SRAM with minimal CPU intervention, maximizing the cycles available for real-time conditioning.
- **Power Domain Configuration:** To ensure logic level compatibility with the sensing array, the internal LDO (VDD_SDIO) must be configured for 3.3V operation. This is achieved by ensuring the MTDI strapping pin is held LOW (0) during the boot sequence, aligning the VDD_SDIO power domain with the VDD3P3_RTC voltage.

MPU-6050 Inertial Measurement Unit

The MPU-6050 provides 6-axis motion tracking, integrating a 3-axis accelerometer and a 3-axis gyroscope. For structural monitoring, the sensor's 1024-byte FIFO buffer is a critical asset; it allows for continuous data accumulation even when the I2C bus is temporarily occupied by the multiplexer switching logic. | Accelerometer Range | Sensitivity (LSB/g) | Primary Application || ----- | ----- | ----- || $\pm 2g$ | 16,384 | Seismic & Modal Bridge Analysis || $\pm 4g$ | 8,192 | General Structural Monitoring || $\pm 8g$ | 4,096 | Industrial Machinery Diagnostics || $\pm 16g$ | 2,048 | High-Impact Event Recording || **I2C Address** | **0x68** | **Base Address** |

I2C Bus Expansion via TCA9548A / PCA9548A

The TCA9548A multiplexer resolves the address conflicts of the sensing array by providing eight bidirectional translating channels. Supporting I2C speeds up to 400 kHz, the multiplexer allows the ESP32 to cycle through sensors by writing a single byte to the control register. A vital engineering detail for long-term deployments on large assets like bridges is the "active LOW" hardware RESET function. This physical reset line allows the ESP32 to recover the I2C bus from stalls caused by Electromagnetic Interference (EMI) or signal degradation over long wire runs without requiring a full system power cycle. While the system supports 400 kHz Fast Mode, actual throughput is carefully tuned to account for the SDA pull-up strength constraints inherent in the ESP32 I2C peripheral. This integrated hardware stack provides the raw high-fidelity data required for advanced digital signal conditioning and noise isolation.

3. Signal Conditioning & Digital Low-Pass Filtering (DLPF)

Isolating relevant structural frequencies from high-frequency environmental and electronic noise is a prerequisite for valid spectral analysis. The Digital Low-Pass Filter (DLPF) within the MPU-6050 serves as the primary hardware-level conditioning stage, ensuring that only the target signal band enters the processing pipeline.

DLPF Register Logic

The filter is configured via the DLPF_CFG2:0 bits in the CONFIG register (Address 1A hex). Enabling the DLPF fundamentally alters the sensor's internal dynamics: while the gyroscope's internal sampling rate drops from 8 kHz to 1 kHz, the accelerometer maintains a fixed 1 kHz internal sample rate. This synchronization at 1 kHz across all axes simplifies the firmware's timing logic and provides a stable base for anti-aliasing.

Filter Characteristics Table

DLPF Setting	Accel Bandwidth (Hz)	Gyro Bandwidth (Hz)	Group Delay (ms)
0	260	256	0.0
1	184	188	2.0
2	94	98	3.0
3	44	42	4.9
4	21	20	8.5
5	10	10	13.8
6	5	5	19.0

From a research perspective, the "Group Delay" is a critical parameter. At the narrowest setting (5 Hz), the 19.0 ms delay must be mathematically accounted for in the timestamping of the data. This delay is particularly significant if the analyzer is integrated into an active

structural damping system, where excessive phase lag in the feedback loop can lead to control instability. These internal sensor configurations define the signal's purity and temporal characteristics before the data is transmitted for final frequency-domain processing.

4. Signal Processing Analysis: Bandwidth vs. FFT Performance

The conversion of physical motion into a meaningful digital frequency spectrum is governed by the Nyquist-Shannon sampling theorem. Proper alignment between the physical sampling rate and the hardware filter bandwidth is the only way to ensure the validity of the resulting FFT.

Nyquist Criteria & Anti-Aliasing

With the accelerometer's internal sampling rate fixed at 1 kHz, the Nyquist frequency is established at 500 Hz. To prevent aliasing—where high-frequency noise is mathematically folded back into the low-frequency spectrum—the DLPF must be used as a front-end anti-aliasing filter. Selecting a bandwidth of 184 Hz or 94 Hz ensures that signal components near or above 500 Hz are significantly attenuated before they are digitized, preserving the integrity of the structural data.

Noise Floor Optimization

Structural assets like bridge decks often exhibit dominant frequencies below 10 Hz. In these scenarios, selecting narrow bandwidths (e.g., 10 Hz or 5 Hz) dramatically improves the Signal-to-Noise Ratio (SNR) by filtering out ambient high-frequency jitter. This optimization of the noise floor is what allows the system to resolve the subtle modal signatures necessary for early-stage fatigue detection.

ESP32 Firmware Sampling Strategy

To maintain FFT accuracy on the host PC, the ESP32 must adhere to a rigid, deterministic sampling interval.

- **1 kHz Synchronization:** The firmware must poll the sensor array at exactly 1 kHz (or integer divisions thereof).
- **The "Stair-Step" Phenomenon:** If the ESP32 polls the I2C bus faster than the internal 1 kHz update rate of the sensor, it will ingest duplicate data points. In the time domain, this manifests as "stair-step" distortion, which creates significant spectral artifacts and "ghost" frequencies in the final FFT. By pinning the I2C task to a dedicated core and utilizing high-resolution hardware timers, the system ensures that every sample represents a unique and accurately timed physical event. The synthesis of dual-core deterministic processing, hardware-level anti-aliasing, and robust I2C bus management establishes this architecture as a high-performance blueprint for the next generation of structural vibration analysis.